Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**K-Agility** Empowering Korean SMEs with verified and quantum-safe, flexible cryptographic solutions in order to be able to compete in global markets			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Information security	*Common Security Technologies*	Software	Software Solutions

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Concept of Proposed Technology	The K-Agility agility framework is an integrated, high-assurance cryptographic system designed to enable South Korean SME's to transition to quantum-safe security without disrupting their existing product architectures. The technology concept is built on a model with 2 layers: 1. The high-assurance verified engine: at the core of the technology is a library of Korean postquantum cryptography (KPQC) algorithms (such as Aimer and HAETAE). Unlike standard software implementations, these modules are validated using computer aided cryptography (CaC). This involves using mathematical logic to formally prove that the code is 100% compliant with the national cryptographic specifications and is free from common vulnerabilities (such as buffer overflows or side-channel leaks). TNO provides the specialized tools and expertise to create these "mathematically perfect" security building blocks. 2. The modular agility management layer: around the verified engine is a software layer, or "agility manager," developed by the SME partner with assistance of TNO. This layer acts as a universal adapter, separating your main software from the security parts so you can swap one out without changing the other. Furthermore, this layer should enable compliance agility, meaning that it is able to effortlessly swap between Korean PQC algorithms for use within Korea and international standards for communication outside Korea. It uses a standard API that allows a system to switch between different encryption methods (for example from classical RSA to KPQC) via a simple configuration change. This means you don't have to rebuild your entire software every time security standards change, a new deployment is set up in a new region with new regulations or a new threat is found. *What is the unique advantage of combining these two technologies?* The real breakthrough is how we combine flexibility with certainty. While most security tools rely on black-box software that you just have to trust, K-Agility agility ensures that every part of your security is mathematically proven to be solid. Instead of relying on the security of cryptographic implementations, we provide formal security guarantees. For a Korean SME, this creates a unique value proposition: a security solution that is both "future-proof" (easy to update), interoperable (able to simultaneously operate in various jurisdictions) and "provably secure" (guaranteed correctness). By making security flexible rather than fixed, this technology removes risk and creates a reliable and high-value asset for South Korean manufacturers and tech companies.
Rationale for Joint R&D Project	The K-Agility project is a partnership where both sides bring a unique strength that the other one lacks. we believe this collaboration is necessary for three reasons: 1. combining high-end mathematics with real-world products: TNO has the specialized skills to use math to prove that security software is 100% correct. This project brings together two different strengths. The SME brings the product and market expertise, while TNO provides the specialized mathematical tools needed for KPQC verification. Together, we create a high-end security solution

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	that would be too time consuming for a single company to build from scratch. CAC has a steep learning curve. Tools are not yet user-friendly. TNO needs a partner who knows the market. The Korean SME understands what customers in South Korea actually need and how to build the final product. Together, we turn high-level research into a tool that businesses can actually use. 2. Making Korean standards global: South Korea has its own national cryptography standards (KPQC). By working together, TNO can apply its international security expertise specifically to these Korean standards. This helps the SME to create a product that is perfect for the Korean market but also meets the high security requirements of global customers in the EU and USA. 3. Sharing the work and the risk: building "quantum-safe" technology is a big challenge. in this project we split the work: TNO handles the complex security "engine," and the SME builds the "software dashboard" that manages it. This allows the SME to get a high-tech product to the international market much faster and with less risk than if they tried to do everything alone. 4. Building a bridge between the Netherlands and South Korea: this project is more than just a one-time deal. It creates a strong link between TNO and the Korean tech industry. it allows TNO to test its ideas in a real industrial setting, and it gives the Korean SME access to world-class European research. this makes both partners stronger and more competitive.
Key Words	1. Post-Quantum Cryptography (PQC) 2. Korean Post-Quantum Cryptography (KPQC) 3. Crypto-Agility 4. Formal Verification 5. High assurance software 6. AIMer / HAETAE 7. Computer aided cryptography (CaC) 8. Digital transformation SME's 9. Quantum ready ICT 10. Cybersecurity strategy

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level **2-3-4**	End of the Project	Level **6-7**
R&D Objectives	The main goal of this project is to develop a finished, "quantum-safe" security framework that allows South Korean SMEs to protect their products with high-end math and total flexibility. By the end of the project we aim to achieve the following four objectives. Objective 1: Creation of a "verified engine" (TRL 4) we will develop a library of Korean national security algorithms (like AImer and HAETAE) that are mathematically proven to be correct. The goal is to produce code that is 100% free from human error and "side-channel" leaks. This provides a solid foundation that SMEs can trust. Objective 2: Development of the "agility manager" (TRL 5) we will build a software dashboard that acts like a universal adapter. This layer will allow a company to swap its security methods (for example, switching from conventional key exchange mechanisms to new KPQC standards) with a simple update and simultaneously operate with different standards. Objective 3: System integration and high-performance testing (TRL 6) we will combine the TNO verified engine with the SME's management layer and test it on real hardware (such as IoT devices). Our target is to ensure that this high-level security does not slow down the device's performance, making it practical for everyday use in the ICT sector. Objective 4: Real-world pilot and demonstration (TRL 7) the final objective is to demonstrate the complete K-Agility agility system in a real-world environment provided by our Korean SME partner. This will prove that the technology is ready for the market and can be used by other South Korean companies to stay competitive and secure.			
R&D Target Specifications	KPI 1 formal verification coverage: Of the resulting implementations at least 90% is covered by formal verification. This way we can ensure highly reliable implementation available for the South-Korean market KPI 2 time to switch: Swapping the encryption algorithm is seamlessly possible without a reboot. By achieving hot-swappable cryptographic systems, we ensure that the product stays protected and operational without any noticeable pause in production. KPI 3 performance: The product is comparable in performance to unverified implementations of similar algorithms, striving to a performance hit of at most 10%. In any case the efficiency of the software is high enough to ensure little to no noticeable latency in high-demand environments. proving that it doesn't make their product too slow or drain the battery is not verified. KPI 4 side-channel: Checking that the product is secured against known methods for side-channel			

	analysis. TNO uses specialized methods to prove that attackers cannot gain access to secret keys by monitoring for example electrical signals or other signals information from the device. KPI value: HIGH
R&D Plan	**The following is a proposal for a plan focusing on both development of a formally verified cryptographic implementation and integrating it into a crypto agile software module; based on the focus of the final project (in conjunction with the final partner) this can be adjusted,** - **Fase 1 (month 1-6).** Choosing protocol(s) to focus on. Start modeling protocols using CaC tooling. - **Fase 2 (month 7-18).** Software development (protocol implementation). Design of crypto agile modules. - **Fase 3 (month 19-30).** Analyze discrepancy between software implementation and model. Align between cryto agile module and developed software final version of model, (verified) software and crypto agility module.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Expertise in computer-aided cryptography. These are the techniques that enable us to formally prove that cryptography meets the security requirements at different levels (design, algorithm, implementation). • Expertise in protocol design, which enables us to develop the required protocols in a way that makes it easy to maintain, updateable and interoperable.
Expected R&R of Korean SME Partner	• Expertise in production-ready software development. • Expertise on setting requirements for hardware development. • Find (local) opportunities for deploying and testing the products delivered as project outcomes. • Integrating the product with a (local) real-world use-case.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The K-Agility agility project will have a significant positive impact on several levels: Economic impact for the SME: - Cost savings: SMEs will save a lot in future maintenance costs because they won't have to rewrite their software every time a new security threat appears. Furthermore, the agility layer enables the Korean SME to seamlessly operate in different jurisdictions. - Global growth: by using TNO's high level verification Korean SME's can prove their products meet the highest international standards. this opens doors to large contracts in the EU and USA, where "quantum-safe" proof is now required. Technological impact for South Korea: - Making the Korean standards work: this project helps South Korea start using its own KPQC security rules quickly. it moves these ideas into real software to protect Korean businesses today. - Leading the world in trust: it puts South Korea at the front of "verified security." This gives Korean technology a global reputation for being the most reliable and trusted on the market. Public and national safety: - Keeping citizens safe: by fixing the security in medical devices, smart cars, and power grids, we protect the daily lives of Korean people from future hacker attacks using quantum computers. - A quantum-ready nation: this project is a big step in making South Korea ready for the future. It ensures the country's private data stays safe for the next 20 to 30 years.

6. References

[1] Korean Post-Quantum Cryptography Competition (2025). SMAUG-T & HAETAE. https://kpqc.cryptolab.co.kr/ *(Last visited: Jan. 21st 2026)*

[2] T. Attema et al. (2024). The PQC Migration Handbook.

[3] Bank for International Settlements (2025). Quantum-readiness for the financial system: a roadmap.

[4] NIST (2025). Considerations for Achieving Cryptographic Agility: Strategies and Practices

[5] NCSC (2024). het crypto-agilitymonster op een bierviltje (Dutch). *https://www.ncsc.nl/blog/het-crypto-agilitymonster-op-een-bierviltje*

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Participating Researcher Profiles

Mr Thomas Attema is a senior researcher in the Applied Cryptography and Quantum Applications department at TNO, and a member of the Cryptology group at CWI (the National Research Institute for Mathematics and Computer Science in the Netherlands). He obtained his PhD in 2023 from the Mathematical Institute of Leiden University. His research focuses on the development of post-quantum cryptography (PQC) and on supporting the migration of digital systems to PQC solutions. In this context, he has co-authored the Dutch PQC Migration Handbook and is an organizer of the international PQC Symposium Series. Thomas has contributed to both the acquisition and execution of several international cryptography projects, including projects funded under the Horizon Europe Programme.

Mr. Bor de Kock is a principal scientist at TNO specialized in Post-Quantum Cryptography, before this he was an associate professor of cryptology at the Norwegian University of Science and Technology in Trondheim. His background is in computer science with a PhD dissertation on Post-Quantum Key Exchange. Bor's research has been funded by the Norwegian Research Council, the German Academic Exchange Service and Horizon Europe.

Mr. Vincent Dunning is a senior scientist in applied cryptography at TNO and external PhD candidate at the University of Twente. His main research interests are migration to Post-Quantum Cryptography and Privacy-Enhancing Technologies. In this role, he has authored both editions of the Post-Quantum Cryptography Migration Handbook, which has been widely adopted in PQC migration roadmaps throughout Europe.

Mr. Michiel Marcus is a senior researcher at TNO specialised in post-quantum cryptography and formal methods. He is a delegate at the ETSI Quantum-Safe Cryptography working group, where he co-authors reports and standards on cryptographic proofs, post-quantum security protocols, and cryptographic agility. His background is in computer science with a Master's degree in cyber security, focused on post-quantum cryptography and quantum computing. The majority of his current research efforts focus on advancing formal verification techniques for cryptography and their integration in industry. Michiel's research has been funded by the Dutch Research Council and various government and industry collaborations.

Mr Frank Westers is a scientist at TNO with a Master's degree in Computational Logic. He currently applies this expertise to advance Computer-Aided Cryptography tools for both software and hardware applications. His research has received funding from an Eureka RD&I Cluster (ITEA) and through partnerships with government and industry collaborators.

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Mr Stefan van den Berg is a researcher at TNO specialised in software testing, formal methods, and post-quantum cryptography. For his double Master's degree in Information Security Technology and Embedded Systems, he implemented a cryptographic library in RISC-V. He focusses on the integration of post-quantum cryptography and secure implementations.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization